

Predicting Student Re-engagement with AI Assistants

AI assistants have become an important part of modern education. However, not all students continue using them after their initial experience. This project aims to develop a supervised machine learning model that predicts whether a student will use an AI assistant again based on factors such as session duration, number of prompts, task type, AI assistance level, discipline, and satisfaction rating. The model can help educational platforms improve user engagement and personalize AI-assisted learning experiences.

Dataset Description

The dataset used in this project is "**AI Assistant Usage in Student Life**", which contains information about how students interact with AI assistants for academic purposes. It includes **10,000 records**, where each record represents an individual AI usage session by a student. The dataset is designed to analyze student's engagement with AI tools and predict whether they are likely to continue using AI assistants in the future.

The dataset contains the following attributes:

- **SessionID:** Unique identifier for each AI session.
- **StudentLevel:** Academic level of the student.
- **Discipline:** Student's field of study.
- **SessionDate:** Date on which the AI assistant was used.
- **SessionLengthMin:** Duration of the AI session in minutes.

- **TotalPrompts:** Number of prompts entered during the session.
- **TaskType:** Purpose of using the AI assistant.
- **AI_AssistanceLevel:** Level of assistance provided by the AI assistant.
- **FinalOutcome:** Outcome of the AI assisted task.
- **UsedAgain:** Indicates whether the student used the AI assistant again (Target Variable).
- **SatisfactionRating:** Student's satisfaction rating after the session.

Data Preprocessing Steps

- Imported the dataset using the Pandas library.
- Checked the dataset dimensions, data types, and column names.
- Verified that the dataset contained no missing values.
- Checked and removed duplicate records.
- Removed the **SessionID** column because it is only an identifier and does not contribute to prediction.
- Converted the **SessionDate** column into datetime format.
- Extracted useful features such as **Month**, **Day**, and **Weekday** from the SessionDate column.
- Removed the original SessionDate column after feature extraction.
- Encoded categorical variables using **Label Encoding**.
- Verified the processed dataset before model training.

Exploratory Data Analysis

Graph 1 – Distribution of UsedAgain

Most students continued using the AI assistant after their first session, indicating high overall engagement with AI tools.

Graph 2 – Satisfaction Rating Distribution

The majority of students gave higher satisfaction ratings, suggesting that AI assistants were helpful for academic tasks.

Graph 3 – Session Length Distribution

Most AI sessions lasted for a moderate duration, showing that students actively interacted with the AI assistant while completing their tasks.

Graph 4 – Task Type Distribution

Coding, homework, and research were among the most common reasons students used AI assistants.

Model Used

Random Forest Classifier was selected as the primary machine learning model because it provides high classification accuracy, handles both numerical and categorical data effectively, and is less prone to overfitting compared to a single Decision Tree.

Training Process

The dataset was divided into two parts using the Train Test Split technique.

- Training Data: **80%**
- Testing Data: **20%**

The training dataset was used to train the machine learning models, while the testing dataset was used to evaluate their performance on unseen data.

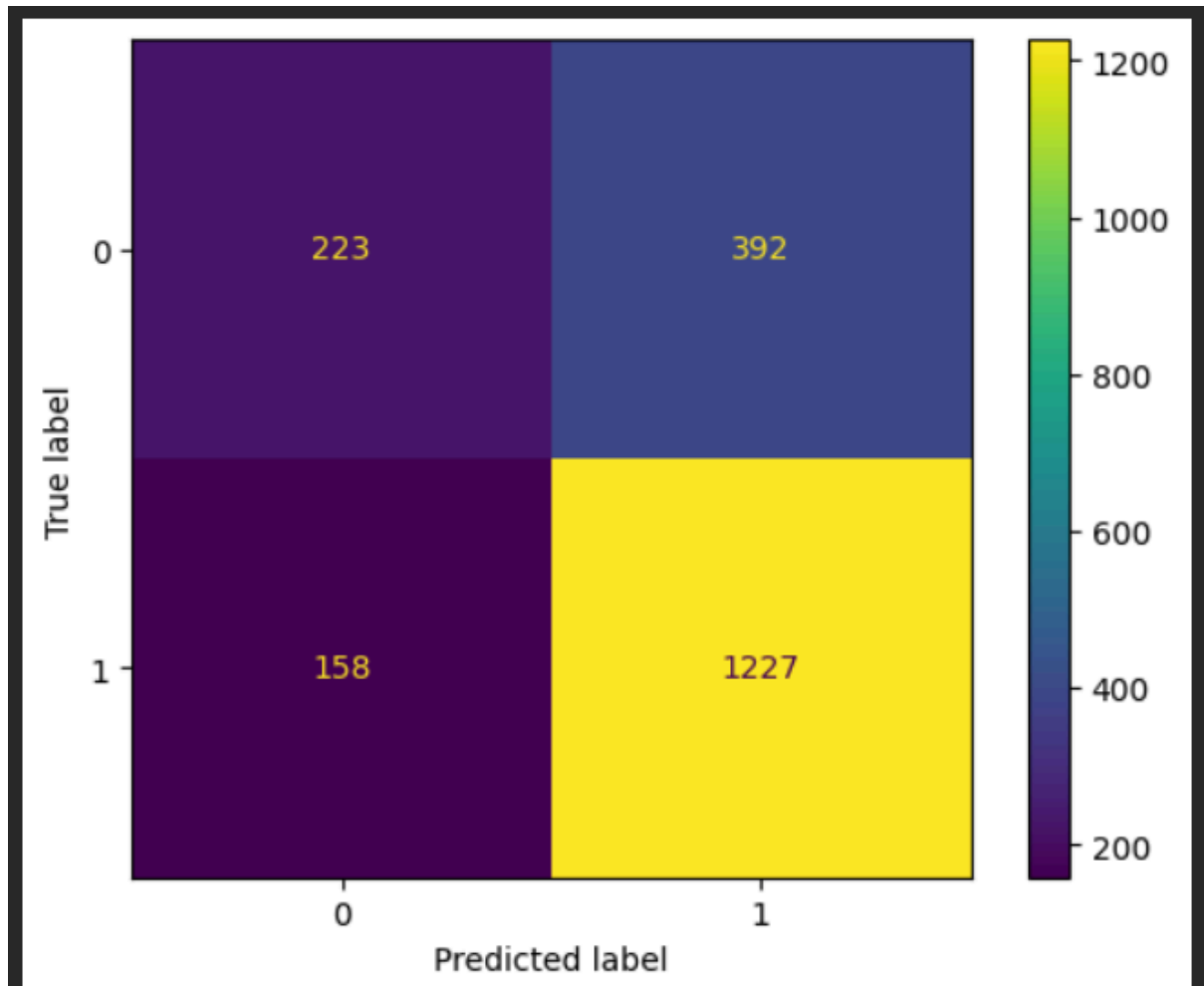
The following steps were followed during training:

1. Feature selection was performed by separating input features and the target variable.
2. The dataset was split into training and testing sets.
3. Random Forest model was trained using the training dataset.
4. Predictions were generated using the testing dataset.
5. The models were evaluated using various classification metrics.

Evaluation Metrics and Results

... Accuracy: 0.725

	precision	recall	f1-score	support
0	0.59	0.36	0.45	615
1	0.76	0.89	0.82	1385
accuracy			0.72	2000
macro avg	0.67	0.62	0.63	2000
weighted avg	0.70	0.72	0.70	2000



Conclusion

This project successfully developed a supervised machine learning model to predict whether a student is likely to use an AI assistant again based on their academic profile, AI usage behavior, and satisfaction level. The dataset was preprocessed by handling duplicate records, removing unnecessary features, converting date information, and encoding categorical variables. Exploratory Data Analysis was performed to understand the distribution of the data and identify important patterns.

References

Kaggle -> [AI Assistant Usage in Student Life](#)